

Phase 4 Causal Transfer Analysis

Scope

- Recipe roots analyzed: runs/transfer_suite/rotating_opponents_holdout_endpoint, runs/transfer_suite/rotating_plus_nemesis_novelty_replay, runs/transfer_suite/rotating_plus_replay_aware_selection.
- Recipes compared: Rotating opponents, Rotating + nemesis + novelty + replay, Rotating + replay-aware selection.
- Baseline recipe for delta metrics: rotating_opponents_holdout_endpoint.
- Condition observations: 90.
- Epoch-to-epoch transition rows: 8910.

Reliability And Replication

- Rotating opponents overall transfer win rate: mean 0.5547 (95% CI 0.4336 to 0.6757; n=10); overall transfer margin: mean 5.9663 (95% CI 1.4844 to 10.4483; n=10).
- Rotating opponents on pursuit / evasion: win rate mean 0.4733 (95% CI 0.2604 to 0.6862; n=10); margin mean -0.4177 (95% CI -4.2727 to 3.4374; n=10).
- Rotating opponents on resource collection / denial: win rate mean 0.404 (95% CI 0.2812 to 0.5268; n=10); margin mean -0.22 (95% CI -2.2085 to 1.7685; n=10).
- Rotating opponents on territory control: win rate mean 0.7867 (95% CI 0.5771 to 0.9963; n=10); margin mean 18.5367 (95% CI 7.7923 to 29.2811; n=10).
- Rotating + nemesis + novelty + replay overall transfer win rate: mean 0.7231 (95% CI 0.7037 to 0.7425; n=10); overall transfer margin: mean 11.6271 (95% CI 9.4761 to 13.7781; n=10).
- Rotating + nemesis + novelty + replay on pursuit / evasion: win rate mean 0.72 (95% CI 0.6656 to 0.7744; n=10); margin mean 4.074 (95% CI 3.0943 to 5.0537; n=10).
- Rotating + nemesis + novelty + replay on resource collection / denial: win rate mean 0.476 (95% CI 0.4348 to 0.5172; n=10); margin mean 1.184 (95% CI 0.723 to 1.645; n=10).
- Rotating + nemesis + novelty + replay on territory control: win rate mean 0.9733 (95% CI 0.952 to 0.9947; n=10); margin mean 29.6233 (95% CI 23.6851 to 35.5615; n=10).
- Rotating + replay-aware selection overall transfer win rate: mean 0.7102 (95% CI 0.6799 to 0.7406; n=10); overall transfer margin: mean 13.0733 (95% CI 10.7589 to 15.3878; n=10).
- Rotating + replay-aware selection on pursuit / evasion: win rate mean 0.7 (95% CI 0.6347 to 0.7653; n=10); margin mean 3.702 (95% CI 2.5161 to 4.8879; n=10).
- Rotating + replay-aware selection on resource collection / denial: win rate mean 0.484 (95% CI 0.45 to 0.518; n=10); margin mean 1.348 (95% CI 0.9859 to 1.7101; n=10).
- Rotating + replay-aware selection on territory control: win rate mean 0.9467 (95% CI 0.904 to 0.9893; n=10); margin mean 34.17 (95% CI 27.5863 to 40.7537; n=10).

Comparison	Scope	Left win	Right win	Win diff	Sign p	Left margin	Right margin	Margin diff
Rotating opponents vs Rotating + nemesis + novelty + replay	pursuit / evasion	0.4733	0.72	0.2467	0.4531	-0.4177	4.074	4.4917
Rotating opponents vs Rotating + nemesis + novelty + replay	resource collection / denial	0.404	0.476	0.072	0.5078	-0.22	1.184	1.404
Rotating opponents vs Rotating + nemesis + novelty + replay	territory control	0.7867	0.9733	0.1867	0.375	18.5367	29.6233	11.0867

Comparison	Scope	Left win	Right win	Win diff	Sign p	Left margin	Right margin	Margin diff
Rotating opponents vs Rotating + nemesis + novelty + replay	overall transfer average	0.5547	0.7231	0.1684	0.0215	5.9663	11.6271	5.6608
Rotating opponents vs Rotating + replay-aware selection	pursuit / evasion	0.4733	0.7	0.2267	0.7266	-0.4177	3.702	4.1197
Rotating opponents vs Rotating + replay-aware selection	resource collection / denial	0.404	0.484	0.08	0.7266	-0.22	1.348	1.568
Rotating opponents vs Rotating + replay-aware selection	territory control	0.7867	0.9467	0.16	1.0	18.5367	34.17	15.6334
Rotating opponents vs Rotating + replay-aware selection	overall transfer average	0.5547	0.7102	0.1556	0.5078	5.9663	13.0733	7.107
Rotating + nemesis + novelty + replay vs Rotating + replay-aware selection	pursuit / evasion	0.72	0.7	-0.02	1.0	4.074	3.702	-0.372
Rotating + nemesis + novelty + replay vs Rotating + replay-aware selection	resource collection / denial	0.476	0.484	0.008	1.0	1.184	1.348	0.164
Rotating + nemesis + novelty + replay vs Rotating + replay-aware selection	territory control	0.9733	0.9467	-0.0267	0.625	29.6233	34.17	4.5467
Rotating + nemesis + novelty + replay vs Rotating + replay-aware selection	overall transfer average	0.7231	0.7102	-0.0129	0.2891	11.6271	13.0733	1.4462

Failure Modes

- Holdout rows are reported as mean win rate / mean score margin across replicates.
- Persistent weak archetypes across all compared recipes: pursuit / evasion / evasion_midline_dodge, resource collection / denial / corner_guard, resource collection / denial / diagonal_probe, resource collection / denial / safe_collector.
- Largest win-rate improvements relative to the baseline appeared in pursuit / evasion / evasion_axis_flip for Rotating + nemesis + novelty + replay (+0.400 win rate), pursuit / evasion / evasion_center_weave for Rotating + nemesis + novelty + replay (+0.300 win rate), territory control / territory_far_corner_claim for Rotating + nemesis + novelty + replay (+0.280 win rate), resource collection / denial / safe_collector for Rotating + replay-aware selection (+0.180 win rate), territory control / territory_diagonal_claim for Rotating + nemesis + novelty + replay (+0.160 win rate).

pursuit / evasion

Holdout archetype	Rotating opponents win/margin	Rotating + nemesis + novelty + replay win/margin	Rotating + replay-aware selection win/margin
evasion_axis_flip	0.6 / 1.848	1.0 / 9.07	0.98 / 8.711
evasion_center_weave	0.52 / 0.532	0.82 / 6.061	0.8 / 5.687
evasion_midline_dodge	0.3 / -3.633	0.34 / -2.909	0.32 / -3.292

resource collection / denial

Holdout archetype	Rotating opponents win/margin	Rotating + nemesis + novelty + replay win/margin	Rotating + replay-aware selection win/margin
center_rush	0.36 / -0.54	0.5 / 0.86	0.46 / 0.98
corner_guard	0.36 / -0.9	0.28 / 0.16	0.36 / 0.38
diagonal_probe	0.36 / -1.6	0.4 / -0.02	0.38 / 0.26
edge_patrol	0.82 / 4.12	0.92 / 5.3	0.92 / 5.42
safe_collector	0.12 / -2.18	0.28 / -0.38	0.3 / -0.3

territory control

Holdout archetype	Rotating opponents win/margin	Rotating + nemesis + novelty + replay win/margin	Rotating + replay-aware selection win/margin
territory_diagonal_claim	0.82 / 22.27	0.98 / 31.34	0.98 / 35.99
territory_far_corner_claim	0.68 / 13.41	0.96 / 27.92	0.86 / 22.8
territory_quadrant_claim	0.86 / 19.93	0.98 / 29.61	1.0 / 43.72

Behavioral Interpretation

- mean_code_novelty is the average lexical code-change magnitude across consecutive epochs.
- mean_descriptor_shift is the average behavioral-descriptor distance across consecutive epochs.
- strategy_switch_count, behavior_cell_count, and specific_adaptation_count are the main proxies for stable adaptive motifs in this report.
- All transitions: novelty vs descriptor shift -> rho 0.2412 (n=8910).
- Accepted transitions only: novelty vs descriptor shift -> rho 0.2503 (n=4902).
- Strongest positive transfer correlates in this dataset: mean_descriptor_shift (0.7598), functional_adaptation_ratio (0.7166).
- Strongest negative transfer correlates in this dataset: superficial_novelty_rate (-0.5923), strategy_switch_rate (-0.4631).

Feature proxy	Outcome	Spearman rho	N
mean_code_novelty	primary_win_rate	0.5735	90
mean_code_novelty	primary_margin	0.6051	90
mean_code_novelty	delta_vs_baseline_win_rate	-0.0589	60
mean_code_novelty	delta_vs_baseline_margin	0.1823	60
mean_descriptor_shift	primary_win_rate	0.7598	90
mean_descriptor_shift	primary_margin	0.7661	90
mean_descriptor_shift	delta_vs_baseline_win_rate	0.0773	60
mean_descriptor_shift	delta_vs_baseline_margin	0.3645	60
functional_adaptation_ratio	primary_win_rate	0.7166	90
functional_adaptation_ratio	primary_margin	0.7198	90
functional_adaptation_ratio	delta_vs_baseline_win_rate	0.0635	60
functional_adaptation_ratio	delta_vs_baseline_margin	0.3467	60
strategy_switch_rate	primary_win_rate	-0.4631	90
strategy_switch_rate	primary_margin	-0.4477	90
strategy_switch_rate	delta_vs_baseline_win_rate	-0.0995	60
strategy_switch_rate	delta_vs_baseline_margin	-0.2427	60
behavior_cell_count	primary_win_rate	-0.255	90
behavior_cell_count	primary_margin	-0.1692	90
behavior_cell_count	delta_vs_baseline_win_rate	-0.0718	60
behavior_cell_count	delta_vs_baseline_margin	-0.0543	60
specific_adaptation_rate	primary_win_rate	0.1086	90
specific_adaptation_rate	primary_margin	0.148	90
specific_adaptation_rate	delta_vs_baseline_win_rate	0.0121	60
specific_adaptation_rate	delta_vs_baseline_margin	0.0983	60
superficial_novelty_rate	primary_win_rate	-0.5923	90
superficial_novelty_rate	primary_margin	-0.6351	90

Feature proxy	Outcome	Spearman rho	N
superficial_novelty_rate	delta_vs_baseline_win_rate	0.0679	60
superficial_novelty_rate	delta_vs_baseline_margin	-0.2133	60
post_loss_novelty_spike_rate	primary_win_rate	0.0873	90
post_loss_novelty_spike_rate	primary_margin	0.0643	90
post_loss_novelty_spike_rate	delta_vs_baseline_win_rate	0.0972	60
post_loss_novelty_spike_rate	delta_vs_baseline_margin	-0.0488	60

Interpretation Guardrails

- These correlations are exploratory. They do not establish causality on their own.
- The behavioral descriptors remain heuristic proxies, but they are closer to functional adaptation than lexical code novelty alone.
- Paired-seed transfer comparisons should be weighted more heavily than unpaired aggregate differences.
- Persistent weak archetypes provide evidence of uneven robustness and remain visible in the report.